
Fernando Lucatelli Nunes

Researcher in Mathematics and Computer Science

PhD in Mathematics, awarded in 2018 (*summa cum laude*)



Webpage: <https://fntlucatelli.github.io/>

arXiv preprints

Mathematics Genealogy

University of Coimbra (CMUC)

- External Researcher at the Centre for Mathematics, University of Coimbra (Portugal)
- Member of the Coimbra Category Theory Group (Portugal)

Date of birth: 31/01/1990

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INTRODUCTION

I am a researcher in pure mathematics and computer science, with a strong track record of research leadership and innovation. After earning my PhD in Mathematics *summa cum laude* in 2018 – through the joint doctoral program of the University of Coimbra and the University of Porto – I have held a range of academic and research positions, including Invited Assistant Professor, Postdoctoral Fellow, Visiting Postdoctoral Researcher, and External Researcher, at leading institutions across Europe, the Americas, and Africa. In these roles, I have been actively involved in research supervision, principal investigator responsibilities, teaching, and academic administration. I have also contributed to strategic initiatives through service on research advisory boards, organization of academic events, participation in hiring committees, and other institutional roles.

Currently, while completing a second PhD in Computer Science at Utrecht University, I also maintain an active research program in mathematics as an External Researcher at the Centre for Mathematics, University of Coimbra. In this role, I work independently on research projects and initiatives, supervise students, and organize seminars. My interdisciplinary expertise allows me to combine theoretical foundations with practical applications, bridging pure mathematics and computer science in a mutually enriching way.

My main area of expertise is the theory and applications of generalized categorical structures in low and higher dimensions, often approached through the lens of two- and three-dimensional universal algebra. My research focuses on developing categorical frameworks that advance knowledge across pure mathematics, computer science, and software engineering. I am passionately committed to interdisciplinary projects, striving to shape the theory of computer science in ways that remain firmly connected to real-world challenges. I particularly enjoy designing theory-based tools that are principled, dependable, and enduring, while also allowing the demands of practice to inform and refine the underlying theory.

My teaching journey began at age sixteen, instructing mathematics in a Brazilian Higher Education Exam preparation course, leading to a diverse career spanning multiple cultural contexts and educational levels, from elementary school to doctoral programs. Beyond traditional classroom teaching, I have engaged in educational outreach, including coaching students for international math competitions and fostering scientific communication in remote areas of Brazil. My commitment to knowledge sharing extends to the international research community, where I have (co-)organized seminars (and events) that have brought together researchers and students across several continents, fostering collaborative environments for scientific exchange.

Throughout my career, I have built a comprehensive record of formal and informal mentorship and leadership, supervising students at all academic levels – from high school to doctoral candidates – as well as postdoctoral researchers. My leadership experience extends to serving on PhD program advisory boards, research advisory boards, and faculty committees, where I have helped design and implement strategic initiatives in both research and education. Drawing on this broad experience, I am committed to developing innovative educational and research programs while contributing towards excellence in their implementation.

UPCOMING POSTDOCTORAL POSITION

2025	Postdoctoral Fellow , School of Computer Science, University of Birmingham, UK.
2025	Postdoctoral Fellow , Department of Information and Computer Sciences, Utrecht University, Netherlands.

POSTDOCTORAL POSITIONS

2020 – present	External Researcher , Centre for Mathematics, University of Coimbra, Portugal.
2023	Fields Research Fellow , Fields Institute for Research in Mathematical Sciences, Canada.
2023	Postdoctoral Fellow , York University, Canada.
2022	Oberwolfach Leibniz Fellow , Mathematisches Forschungsinstitut Oberwolfach, Germany.
2018 – 2021	(Postdoctoral) Junior Research Fellow , Department of Mathematics, University of Coimbra, Portugal.
2018 – 2019	Invited Assistant Professor , Department of Mathematics, University of Coimbra, Portugal.
2018	Postdoctoral Fellow , Centre for Mathematics, University of Coimbra, Portugal.

Visiting Positions

2018, 2023, 2024, 2025	Postdoctoral Visiting Researcher , UCLouvain, Belgium.
2025	Postdoctoral Visiting Researcher , IMPA, Rio de Janeiro, Brazil.
2019, 2025	Postdoctoral Visiting Researcher , University of Cape Town, South Africa.
2019, 2025	Postdoctoral Visiting Researcher , Stellenbosch University, South Africa.

OTHER POSITIONS

2020 – 2025	Promovendus/Doctoral Researcher , Utrecht University, Netherlands.
2012 – 2018	Doctoral Researcher/PhD candidate , University of Coimbra and University of Porto, Portugal.
2006 – 2008	Secondary Math Teacher , Curso Dínatos, Brasília, Brazil.
2005 – 2006	Pre-cadet/ “Aluno” , EPCAR, Brazilian Air Force, Brazil.

PhD IN MATHEMATICS (awarded in 2018)

10/2012 – 01/2018

University of Coimbra, Ph.D. in Mathematics

- Thesis: Pseudomonads and Descent (awarded summa cum laude)
- Supervisor: Professor Doctor Maria Manuel Clementino
- Keywords of the contributions: biadjoint triangle theorems, freely generated categorical structures, theory of presentation of categorical structures, coherence in low and higher dimensional category theory, Grothendieck descent theory, Galois theories, generalized categorical structures, two-dimensional monad theory, fundamental aspects of two-dimensional category theory, theory of low and higher dimensional weak limits.

PhD IN COMPUTER SCIENCE (about to finish in 2025)

11/2020 – present

Utrecht University, Ph.D. in Computer Science

- Thesis: Freely generated categorical structures and automatic differentiation
- Supervisors: Professor Doctor Gabriele Keller and Doctor Matthijs Vákár
- Keywords of the contributions: programming language denotational semantics, automatic differentiation, software correctness, program transformations, categorical semantics, logical relations, type systems, dependent types, algebraic effects.

(Selected) Participation in Projects, Grants and Fellowships

2025 – present	ERC starting Grant FoRECAST, Formalised Reasoning about Expectations: Composable, Automated, Speedy, Trustworthy. • PI/Grant holder: Matthijs Vákár. • My role: Working in the formal methods team, especially when devising principled algorithms, employing categorical structures. My role includes the (co)supervision of PhD students.
2025 – present	DFG-Project HOMBRe, Higher-Order Monad-based Programming and Reasoning. • PI/Grant holder: Sergey Goncharov. • My role: Postdoctoral Fellow, working in the main topics of the project.
2021 – 2024	NWO Veni grant number VI.Veni.202.124, Correct Expressive Differential Programming: Laying the Mathematical Foundations for Tomorrow’s Machine Learning. • PI/Grant holder: Matthijs Vákár • My role: Led the work on Iterative CHAD, working on the formal methods endeavors of the project. Also, proved dual numbers (forward and reverse mode) AD for languages featuring recursive types
2020 – 2022	Marie Skłodowska-Curie grant, Principled AD program transformations • PI/Grant holder: Matthijs Vákár. • My role: Worked on extending reverse-mode AD (CHAD) program transformations correctly to expressive total languages.
2022 – 2022	Oberwolfach Leibniz Fellowship, Grothendieck Descent Theory, Galois Theory and Presentations of Categorical Structures • PI: Fernando Lucatelli Nunes • Participants: Fernando Lucatelli Nunes, Maria Manuel Clementino, Rui Prezado, Lurdes Sousa, Matthijs Vákár.
2023 – 2023	Fields Research Fellowship, Grothendieck Descent Theory and Fibrations, and Factorizations • PI: Fernando Lucatelli Nunes • Participants: Fernando Lucatelli Nunes, Walter Tholen.
2011 – 2011	Project: Nonlinear Analysis and Applications (CNPq), Group Project (University of Brasilia), financed by CNPq • Title: Nonlinear Analysis and Applications • My role: Undergrad student, leading the project “Topological Dynamics and applications in Number Theory”, and wrote two monographs on it.

SOME SCHOLARSHIPS AND STUDENT/DOCTORAL FELLOWSHIPS

2020 – 2025	PhD Fellowship (Promovendus) , Department of Information and Computing Sciences, Utrecht University, the Netherlands.
2013 – 2017	PhD Fellowship , National Council for Scientific and Technological Development, CNPq, Brazil.
2012 – 2013	Doctoral Research Grant , Centre for Mathematics of University of Coimbra, Portugal.
2011	Scholarship for 1st year of Master's studies , IMPA, Brazil.
2009 – 2011	Scientific Initiation/Undergraduate research fellowships , National Council for Scientific and Technological Development, CNPq, Brazil. • Grant for Scientific Initiation under the Project Nonlinear Analysis and Applications. Introduction to Topological Dynamics and Applications to Number Theory. • Grant for Scientific Initiation on Lie Theory and Atomic Theory.
2009 – 2012	Several teaching grants , University of Brasília, Brazil.
2005	Aluno , EPCAR, Brazilian Air Force, Brazil

Programming Languages and Technical Expertise

My research in programming languages and program analysis has equipped me with strong adaptability across diverse software technologies and programming paradigms. I have supervised projects involving functional programming languages (Haskell, OCaml), mainstream multi-paradigm languages (Python, C#), JavaScript/TypeScript, and a variety of development frameworks.

NATURAL LANGUAGES

Portuguese and English

Studying/interested in learning: Swedish, Dutch and French

RESEARCH INTERESTS

Expertise and research interests: Mathematics, Computer Science and Software Engineering, including: (1) the theory and applications of low and higher dimensional generalized (multi)categorical structures; (2) mathematical logic, (3) formal methods in computer science, (4) programming languages, (5) static program analysis, (6) functional programming, (7) algebraic topology and geometry, (8) homotopy theory, (9) low and higher dimensional universal algebra.

RESEARCH EXPERIENCE

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|----------------|--|
| 2020 – present | <p>External Researcher, Centre for Mathematics, University of Coimbra, Portugal.</p> <ul style="list-style-type: none">• I have undertaken single-authored research projects spanning a broad range of topics in the theory and applications of low and higher dimensional (generalize) (multi)categorical structures, including most of the topics listed in my research interests.• Collaborating with Marcelo Fiore, Tim Van der Linden, Walter Tholen, Lurdes Sousa, Maria Manuel Clementino, Rui Prezado, João Nogueira, and others.• Supervising students (finished the supervision of Rui Prezado’s PhD Thesis in 2024).• Member of the Portuguese Category Theory Group.• Member/Collaborator of the research group Algebra, Logic and Topology. |
| 2025 | <p>Postdoctoral Visiting Researcher, University of Cape Town, South Africa.</p> <ul style="list-style-type: none">• Intend to work in Grothendieck descent theory for topological manifolds! |
| 2025 | <p>Postdoctoral Visiting Researcher, Stellenbosch University, South Africa.</p> <ul style="list-style-type: none">• Intend to work in collaboration with Graham Richard Manuell in a project concerning Cauchy completion of categories. |
| 2020 – 2025 | <p>Doctoral Researcher in Computer Science, Department of Information and Computing Sciences, Utrecht University, Netherlands.</p> <ul style="list-style-type: none">• Collaborative research on automatic differentiation for expressive programming languages, with a focus on correctness proofs and implementation challenges, in collaboration with Matthijs Vákár, Tom Smeding, and Gordon Plotkin.• Research in progress on general aspects of denotational semantics of programming languages with polymorphism, including algebraic effects, (co)inductive types, existential types and recursive types.• Research in progress on constructs for proof assistants allowing non-termination. Algebraic effects in dependently typed languages, recursion and iteration.• Research in progress on open semantic logical relations techniques for expressive programming languages.• Research in progress on program transformations, including closure conversion, automatic differentiation, predicate transformers, symbolic program analysis, among other program transformations in general-purpose or domain specific languages.• Supervisors: Professor Gabriele Keller and Dr. Matthijs Imre László Vákár. |
| 2025 | <p>Postdoctoral Visiting Researcher, IMPA, Rio de Janeiro, Brazil.</p> |

- Category theory at IMPA!
 - Delivered a talk entitled “Biadjoint Triangles and Coherence” on the 16th of May, where I could convince the audience that a “purely categorical result” can be interesting for people working in other subjects related to Representation Theory, Poisson Geometry, and Vertex Algebras;
 - Developed collaboration opportunities, including research collaboration in subjects connected to algebraic topology and homotopy theory;
 - Talked to the students with the aim of finding good candidates to the open position of PhD student within the context of the ERC starting grant project FoRECAST (led by Matthij Vákár and co-supervised by me);
 - Started to co-supervise a Master Student.
- 2025 **Postdoctoral Visiting Researcher**, UCLouvain, Belgium.
- Conducting research in collaboration with Tim Van der Linden and Marcelo Fiore.
 - Working in a project called “Infinitely Monoidal Categories”.
- 2024 **Postdoctoral Visiting Researcher**, UCLouvain, Belgium.
- Conducting research in collaboration with Rui Prezado and Matthijs Vákár.
 - Explored Freely generated categorical structures, related with my work on programming language semantics: freely generated (doubly-)infinitary extensive categories.
 - Wrote the paper “Free extensivity via distributivity” (published).
 - Participation in UCLouvain-ULB-VUB Category Theory Seminar.
- 2023 **Fields Research Fellow**, Fields Institute for Research in Mathematical Sciences, Canada.
- Conducting research in collaboration with Walter Tholen on Pure Mathematics.
 - Exploring Categorical Aspects of Grothendieck Descent Theory.
- 2023 **Postdoctoral Researcher**, York University, Canada.
- Conducting research in collaboration with Walter Tholen on Pure Mathematics.
 - Specifically investigating Grothendieck Fibrations and Factorization Systems.
- 2023 **Postdoctoral Visiting Researcher**, UCLouvain, Belgium.
- Worked closely with Rui Prezado, my PhD student, to explore Generalized Multicategories and the dichotomy between internal vs. enriched categorical structure (motivated by some ideas of my PhD thesis); and effective descent morphisms in this setting. This resulted on the paper “Generalized multicategories: change-of-base, embedding, and descent”.
- 2022 **Oberwolfach Leibniz Fellow**, Mathematisches Forschungsinstitut Oberwolfach, Germany.

- Collaborated on a denotational correctness proof of forward and reverse mode dual numbers AD for realistic functional languages in the ML-family, resulting in the paper “Automatic Differentiation for ML-family languages: correctness via logical relations” (co-authored with Matthijs Vákár). The efficient implementation of such AD macro was later implemented by Tom Smeding and Matthijs Vákár.
- Characterized fully faithful (enriched) functors that are also co-fully faithful and developed techniques for studying descent morphisms of fibrations not satisfying the Beck-Chevalley condition, resulting in the preprint “Cauchy Completeness, Lax Epimorphisms, and Effective Descent for Split Fibrations” (co-authored with Rui Prezado and Lurdes Sousa).
- Collaborated with Matthijs Vákár to create simple-enough open semantic logical relations techniques for programming languages featuring ML-polymorphism and recursive types.
- Collaborated to finish a work providing an orthogonal factorization system in the category of categories, with the left side of the factorization given by the co-fully faithful functors, resulting in publication of “On lax epimorphisms and the associated factorization” (co-authored with Lurdes Sousa).
- Collaborated with Maria Manuel Clementino to establish fundamental properties (e.g. topologicity, descent and exponentiability) of the category of free split fibrations. This resulted in the paper “Lax comma categories: cartesian closedness, extensivity, topologicity, and descent”.
- Collaborated with Maria Manuel Clementino on descent and exponentiability in lax comma categories.
- Collaborated with Lurdes Sousa on lax epimorphisms and factorization systems in the category of categories.

2018 – 2021

Research Fellow, Department of Mathematics, University of Coimbra, Portugal.

- Proved that morphisms that forget descent data create absolute Kan extensions. Implications in Grothendieck descent theory and monadicity theorem. (ref: “Descent Data and Absolute Kan Extensions”).
- Introduced lax 2-dimensional analogue of (co)image factorization in 2-categories, the lax semantic factorization. Proved coincidence with (co)Eilenberg-Moore factorization for morphisms with adjoints. As a consequence, demonstrated that right adjoint functors are monadic if and only if they are effective faithful morphisms. (from “Semantic Factorization and Descent”).
- Member of the research group “Algebra, Logic and Topology” at the Centre for Mathematics, University of Coimbra.

2019

Postdoctoral Visiting Researcher, University of Cape Town, South Africa.

- Collaborated with Maria Manuel Clementino on developments towards 2-dimensional aspects of Janelidze-Galois Theory.

2018

Postdoctoral Fellow, Centre for Mathematics, University of Coimbra, Portugal.

- Provided a unified perspective on non-canonical isomorphisms through the lens of two-dimensional monad theory, as presented in the publication “Pseudocalgebras and non-canonical isomorphisms”.
- Fellow at the Algebra, Logic and Topology research group.

2018

Postdoctoral Visiting Researcher, Institut de Recherche en Mathématique et Physique, Université catholique de Louvain. Louvain-la-Neuve, Belgium.

2012 – 2018

- Started my project on seeing descent data as a generalization of (co)algebraic structure, resulting in multiple works including “Descent Data and Absolute Kan Extensions” and “Semantic Factorization and Descent” (some of them yet unpublished).
- Collaborated with Maria Manuel Clementino on the development of a suitable notion of admissible 2-functor, compatible with her previous work on lax orthogonal factorization systems.

Doctoral Researcher, University of Coimbra and University of Porto, Portugal.

- Contributed to the foundations (including topological and combinatorial aspects) of the theory of presentation of low dimensional categories. Part of this work can be found in “Freely generated n-categories, coinserters and presentations of low dimensional categories”.
- Contributed to the 2-dimensional foundations of Grothendieck descent theory, including proofs of classical results in terms of commutativity of bilimits (published in “Pseudo-Kan Extensions and Descent Theory”).
- Introduced the notion of (Pointwise) pseudo-Kan extension and pseudo-end, as published in “Pseudo-Kan Extensions and Descent Theory” and “On Biadjoint Triangles”.
- Proved 2-dimensional adjoint triangle theorems, as published in “On Biadjoint Triangles” and “On lifting of biadjoints and lax algebras”.
- Utilized these results to derive a characterization of the 2-monads that allow coherences results of the type “strict”/cofibrant replacements of pseudoalgebras.

- Lucatelli Nunes, F., Vákár, M.: **Automatic differentiation for ML-family languages: Correctness via logical relations**. Mathematical Structures in Computer Science. 34, No. 8, 747–806 (60 pages).
- Prezado, R., Lucatelli Nunes, F.: **Generalized multicategories: change-of-base, embedding, and descent**. Appl. Categ. Struct. Volume 32, 35. 2024. (89 pages).
- Lucatelli Nunes, F., Prezado, R., Vákár, M.: **Free extensivity via distributivity**. Portugaliae Mathematica, 82 (2025), no. 1/2, pp. 177–204. (28 pages).
- Clementino, M.M., Lucatelli Nunes, F., Prezado, R.: **Lax comma categories: cartesian closedness, extensivity, topologicity, and descent**. Theory Appl. Categ. 41, 516–530. 2024. (15 pages).
- Clementino, M.M., Lucatelli Nunes, F.: **Lax comma 2-categories and admissible 2-functors**. Theory Appl. Categ. 40, 180–226. 2024. (47 pages).
- Lucatelli Nunes, F., Vákár, M.: **CHAD for Expressive Total Languages**. Math. Struct. Comput. Sci. 33, No. 4–5, 311–426. 2023. (116 pages).
- Clementino, M.M., Lucatelli Nunes, F.: **Lax comma categories of ordered sets**. Quaest. Math. 46, Part Suppl. 1, 145–159. 2023. (14 pages).
- Lucatelli Nunes, F., Prezado, R., Sousa, L.: **Cauchy Completeness, Lax Epimorphisms, and Effective Descent for Split Fibrations**. Bull. Belg. Math. Soc. - Simon Stevin 30, No. 1, 130–139. 2023. (10 pages).
- Prezado, R., Lucatelli Nunes, F.: **Descent for internal multicategory functors**. Appl. Categ. Struct. 31, 11. 2023. (18 pages).
- Lucatelli Nunes, F.: **Semantic factorization and descent**. Appl. Categ. Struct. 30, No. 6, 1393–1433. 2022. (41 pages).
- Lucatelli Nunes, F., Sousa, L.: **On lax epimorphisms and the associated factorization**. J. Pure Appl. Algebra 226, No. 12, 27 p. 2022. (27 pages).
- Lucatelli Nunes, F.: **Descent data and absolute Kan extensions**. Theory Appl. Categ. 37, 530–561. 2021. (32 pages).
- Lucatelli Nunes, F.: **Pseudoalgebras and non-canonical isomorphisms**. Appl. Categ. Struct. 27, No. 1, 55–63. 2019. (9 pages).
- Lucatelli Nunes, F.: **On lifting of biadjoints and lax algebras**. Categ. Gen. Algebr. Struct. Appl. 9, No. 1, 29–58. 2018. (30 pages).
- Lucatelli Nunes, F.: **Pseudo-Kan Extensions and Descent Theory**. Theory Appl. Categ. 33, 390–444. 2018. (35 pages).
- Lucatelli Nunes, F.: **On biadjoint triangles**. Theory Appl. Categ. 31, 217–256. 2016. (40 pages).
- Lucatelli Nunes, F.: **Espaços não reversíveis**. (Non-reversible spaces). Revista Matemática Universitária, v. 48, n. 49, p. 22–26. 2012. (5 pages).

PREPRINTS

- Lucatelli Nunes, F., Plotkin, G., Vákár, M.: **Unraveling the iterative CHAD**. arXiv:2505.15002. 2025. (57 pages).
- Lucatelli Nunes, F., Prezado, R.: **Functors Preserving Effective Descent Morphisms**. arXiv:2410.22876. 2024. (17 pages).
- Lucatelli Nunes, F., Vákár, M.: **Monoidal closure of Grothendieck constructions via Σ -tractable monoidal structures and Dialectica formulas**. arXiv:2405.07724. 2024. (28 pages).
- Lucatelli Nunes, F., Vákár, M.: **Free doubly-infinitary distributive categories are cartesian closed**. arXiv:2403.10447. 2024. (16 pages).
- Lucatelli Nunes, F.: **Freely generated n-categories, coinserters and presentations of low dimensional categories**. arXiv:1704.04474. 2017. (56 pages).

WORKSHOP AND OTHER PUBLICATIONS

- Lucatelli Nunes, F., Plotkin, G., Vákár, M.: **Homomorphic Reverse Differentiation of Iteration**. LAFI 2024 (Tenth Workshop on Languages for Inference at POPL, 2024). 2024. (2 pages).
- Lucatelli Nunes, F., Vákár, M.: **Logical Relations for Partial Features and Automatic Differentiation Correctness**. arXiv:2210.08530. 2022. and Oberwolfach Preprints, OWP-2023-09. 2023. (21 pages).
- Lucatelli Nunes, F.: **Pseudomonads and Descent**. PhD Thesis, University of Coimbra. 2017. (209 pages).

UNDERGRADUATE AND GRADUATE MONOGRAPHS

- Lucatelli Nunes, F.: **Basic Concepts of Topology for Topological Dynamics (in Portuguese)**. 2009. (116 pages).
- Lucatelli Nunes, F.: **Basic Topological Dynamics and Basic Applications to Number Theory (in Portuguese)**. 2009. (57 pages).

PHD STUDENTS (THESIS SUPERVISION)

- 2026 – present **Utrecht University**, Department of Information and Computing Science
- Student: Diogo Simm.
 - Thesis title: TBD
 - Keywords: programming language semantics, category theory
 - Co-supervising with Dr. Matthijs Vákár.
 - Mathematics Genealogy
 - Work under the ERC starting Grant FoRECAST.
- 2020 – 2024 **University of Coimbra**, Joint University of Coimbra-University of Porto
PhD program in Mathematics, Portugal
- Student: Rui Rodrigues de Abreu Fernandes Prezado.
 - Thesis title: Some Aspects of Descent Theory and Applications.
 - Keywords: generalized (multi)categories; internal categories; effective descent morphisms; Grothendieck descent theory.
 - Co-supervised with Professor Dr. Maria Manuel Clementino.
 - Mathematics Genealogy
 - Rui Prezado was awarded his PhD with *summa cum laude* in Jan/2024.

MASTER'S STUDENTS (THESIS SUPERVISION)

- 2025 **IMPA, Rio de Janeiro, Brazil**, Master's Program in Pure Mathematics
- Student: Diogo Simmm.
 - Co-supervised with Prof. Henrique Bursztyn and Dr. Matthijs Vákár
 - Keywords: Programming Language Semantics, Automatic Differentiation, Category Theory

MASTER'S STUDENTS (RESEARCH PROJECTS)

- 2025 **Utrecht University, the Netherlands**, Scientific Methods in Computing Science, Programming Languages
- Students: Thomas van Maaren, Job Vonk, and Daan van Westerlaak.
 - Paper: Calculating the Symbolic Derivative using Automatic Differentiation
 - Keywords: Programming Language Theory, Programming Language Design, Categorical Semantics, Automatic Differentiation.
- 2025 **Utrecht University, the Netherlands**, Scientific Methods in Computing Science, Programming Languages
- Students: Hans Lous, Sem van Nieuwanhuizen, Sebastiaan Koppen, and Martijn Voorwinden.
 - Paper: Lambda Set Specialization for Haskell
 - Keywords: Compilers, Program Transformations, Programming Language Design, Categorical Semantics, Defunctionalization.
- 2025 **Utrecht University, the Netherlands**, Scientific Methods in Computing Science, Programming Languages

- Students: Meike van Engelenburg, Tjalle Schoonderwoerd, and Martijn Voordouw.
- Paper: Dual-numbers automatic differentiation with inductive types
- Keywords: Programming Language Theory, Programming Language Design, Categorical Semantics, Automatic Differentiation.

UNDERGRADUATE STUDENTS (FINAL THESIS SUPERVISION)

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| 2025 | <p>Utrecht University, Bachelor's Artificial Intelligence thesis</p> <ul style="list-style-type: none"> • Student: <u>Sander Santema</u>. • Topic: A simple defunctionalization local transformation for simply typed lambda calculus. • Keywords: Categorical semantics; Program Transformation; Defunctionalization, Haskell, Generalized Algebraic Data Types, Functional Programming. |
| 2023 | <p>Utrecht University, the Netherlands, Software Final Project of the BSc program in Computer Science</p> <ul style="list-style-type: none"> • Students: <u>Cérine Toubert</u>, <u>Oscar Mensink</u>, <u>Kasper Peeters</u>, <u>Nigel Strachan</u>, <u>Joost Laan</u>, <u>Jippe Hoogeveen</u>, <u>Gust Verhees</u>, <u>Nasr Antar</u>, and <u>Artur Sałek</u>. • Developing a software dedicated to find ideal matches for collaboration in a company or in the workplace in general, according to interactions, rates and other weights/values. • Client: Assistant Professor Ioanna Lykourantzou • Keywords: optimization algorithms; software development; web development. |

UNDERGRADUATE STUDENTS (RESEARCH/SOFTWARE PROJECTS)

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| 2021-2022 | <p>Utrecht University, the Netherlands, Game Technology</p> <ul style="list-style-type: none"> • A group of nine students, divided into three groups of three. • Developing a game in C#. |
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HIRING COMMITTEES

- 2025 **FoRECAST**, Project “Formalised Reasoning about Expectations: Composable, Automated, Speedy, Trustworthy”
- Member of the hiring committee.
 - Hiring three PhD candidates under the ERC Starting Grant project “Formalised Reasoning about Expectations: Composable, Automated, Speedy, Trustworthy”

PhD COMMITTEES (JURY MEMBER)

- 2024 **PhD defense, University of Coimbra**, Joint University of Coimbra-University of Porto PhD program in Mathematics, Portugal
- Student: **Rui Rodrigues de Abreu Fernandes Prezado**.
 - Topic: Some Aspects of Descent Theory and Applications.
 - **Role:** Supervisor and examiner.
- 2023 **PhD defense, University of Coimbra**, Joint University of Coimbra-University of Porto PhD program in Mathematics, Portugal
- Student: **Carlos Miguel Alves Fitas**.
 - Topic: On lax idempotent monads in topology.
 - **Role:** Main examiner, main opponent.

EVALUATION PHD PROJECT COMMITTEES (JURY MEMBER)

- 2019 **PhD project defense**, Joint University of Coimbra-University of Porto PhD program in Mathematics, Portugal
- Candidate: **Carlos Miguel Alves Fitas**.
 - Topic: On lax idempotent monads in topology.
 - **Role:** Main examiner, main opponent.
- 2019 **PhD project defense**, Joint University of Coimbra-University of Porto PhD program in Mathematics, Portugal
- Candidate: **Rui Rodrigues de Abreu Fernandes Prezado**.
 - Topic: Some Aspects of Descent Theory and Applications.
 - **Role:** Supervisor and examiner.

OTHER EVALUATION COMMITTEES

- 2025 **Final Thesis Proposal**, Bachelor in Artificial Intelligence, Utrecht University, the Netherlands
- Candidate: **Vita Blom**.
 - Topic: Generative AI as a Study Tool in Higher Education: Exploring Its Role in Bloom’s Cognitive Learning Levels.
 - **Role:** Second Reader.

SELECTED TALKS (conferences)

Leiden University, The Netherlands, 2024	109th Peripatetic Seminar on Sheaves and Logic , The semantic lax descent factorization of a functor.
University of Santiago de Compostela, Spain, 2024	International Conference in Category Theory 2024 , Doubly-infinitary distributive categories
University of Coimbra, Portugal, 2024	SUMTOPO 2024, Topology and Logic in Computer Science , Homomorphic reverse differentiation for partial features
University of Coimbra, Portugal, 2024	38th Summer Conference on Topology and its Applications , Free distributive and extensive categories
University of Coimbra, Portugal, 2023	XIV Portuguese Category Seminar , Freely generated bicartesian categories and exponentials
Université catholique de Louvain, Belgium, 2023	International Conference in Category Theory 2023 , Spanning the tale of “Monades et descente”
University of Coimbra, Portugal, 2022	XIII Portuguese Category Seminar , CHAD: elementary categorical aspects of automatic differentiation
University of Primorska, Slovenia, 2021	8th European Congress of Mathematics , Eilenberg-Moore, Kleisli and descent factorizations.
University of Coimbra, Portugal, 2019	XII Portuguese Category Seminar , Lax comma 2-categories.
University of Edinburgh, Scotland, 2019	International Conference on Category Theory 2019 , Descent and Monadicity.
Stellenbosch University, SA, 2019	Categorical Structures in Algebra, Topology and Logic , Eilenberg-Moore, Kleisli and descent factorization.
Masaryk University, Brno, Czech Republic, 2018	103rd Peripatetic Seminar on Sheaves and Logic , Biadjoint Triangles and Applications.
Universidade de Santiago de Compostela, Spain, 2018	102nd Peripatetic Seminar on Sheaves and Logic , Non-canonical isomorphisms.
Université catholique de Louvain, Belgium, 2017	5th Workshop on Categorical Methods in Non-Abelian Algebra , Biadjoint Triangles.
University of Coimbra, Portugal, 2017	XI Portuguese Category Seminar , Freely generated n-categories and comonsters.
University of Coimbra, Portugal, 2016	4th Workshop on Categorical Methods in Non-Abelian Algebra , Bilimits Commutativity and Descent
University of Aveiro, Portugal, 2015	International Conference on Category Theory 2019 , Kan extensions and descent theory.

SPECIAL TALKS

Lorentz Center, Leiden, Netherlands, 2022	New Challenges in Programming Language Semantics (INVITED) , Categorical Approach to Semantic Logical Relations (for partial features and recursive types).
University of Azores, 2018	International Conference on Category Theory (PLENARY) , Aspects of Descent via Bilimits.

(Invited 1-Hour Seminar) TALKS

IMPA, Brazil, 2025	Algebra Seminar , Biadjoint Triangles and Coherence
University of Coimbra, Portugal, 2024	Algebra, Logic and Topology Seminar , Unraveling the iterative CHAD.
University of Coimbra, Portugal, 2022	Algebra, Logic and Topology Seminar , Descent for Split Fibrations.
Utrecht University, Netherlands, 2021	Software Technology Group Seminar , Automatic differentiation of expressive total functional languages.
University of Coimbra, Portugal, 2019	Seminar of the Category Theory Group at the University of Coimbra , On monadicity and descent.
University of Coimbra, Portugal, 2019	Algebra, Logic and Topology Seminar , Functorial Semantics and Descent II.
University of Coimbra, Portugal, 2018	Algebra, Logic and Topology Seminar , Functorial Semantics and Descent.
University of Coimbra, Portugal, 2018	Algebra, Logic and Topology Seminar , Biadjoint Triangles and Applications.
University of Coimbra, Portugal, 2018	Algebra, Logic and Topology Seminar , Towards semi left exactness and Janelidze-Galois within the lax idempotent context.
University of Coimbra, Portugal, 2016	Algebra, Logic and Topology Seminar , The deficiency of categories.
University of Coimbra, Portugal, 2015	Algebra, Logic and Topology Seminar , Biadjoint triangles, descent and coherence.
Universidade do Porto, Portugal, 2016	Research Seminar Program , Kan construction of adjunctions.
University of Coimbra, Portugal, 2016	Research Seminar Program , Homotopy Excision.
University of Coimbra, Portugal, 2013	Joint PhD Program UC-UP Seminar , Every prespectrum represents a homology theory.
Universidade de Brasília, Brazil, 2010	Math Seminar , Topological Dynamics and Applications in Number Theory (in Portuguese).
Universidade de Brasília, Brazil, 2010	Math Seminar , Dynamical Proof of the Van der Waerden Theorem (in Portuguese).

WORKSHOP TALKS

LAFI, London, UK	LAFI 2024 (Tenth Workshop on Languages for Inference at POPL, 2024) , Homomorphic Reverse Differentiation of Iteration.
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OUTREACH ACTIVITIES/SCIENTIFIC DIVULGATION/POPULARIZING SCIENCE

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|-------------|---|
| 2017 – 2020 | <p>Delfos Project, Department of Mathematics, University of Coimbra, Portugal.</p> <ul style="list-style-type: none"> • Collaborated to the Delfos Project, a Mathematics education program for secondary school students preparing for International Math Olympiads. • Portuguese IMO commissioner in 2020. |
| 2008 – 2012 | <p>Scientific Initiation, Scientific Initiation, CNPq, Brazil.</p> <ul style="list-style-type: none"> • As an undergraduate member of the group “Teoria de Lie e Dinâmica”, contributed to the blog https://liedinamica.wordpress.com/, and produced three monographs in Portuguese, covering the subjects of “Basic Lie Groups”, “Basic Concepts of Topology for Topological Dynamics”, and “Basic Topological Dynamics and Basic Applications to Number Theory”. • Published a paper in “Revista Matemática Universitária”, titled “Non-reversible spaces” in Portuguese. |
| 2011 – 2012 | <p>Brazilian Society for Scientific Progress, Presented multiple popular science mathematics posters (in Portuguese) at the Brazilian Society for Scientific Progress annual conferences (examples below), all of which were published in their annals (all in Portuguese).</p> <ul style="list-style-type: none"> • Game Theory: Nash Equilibrium and Evolutionary Games. • Modern Homotopy Theory: Model Categories. • Category Theory: Adjunctions, Colimits and Limits. • An invitation to topology: topological proofs of classical results. • Topological Dynamics and Applications in Number Theory. |

TEACHING EXPERIENCE

2020 – present	AiO/TA (Lecturer), Department of Information and Computing Sciences, Utrecht University, Netherlands. • Tutorials of Functional Programming course. • Awarded “Teaching award” (Distinguished Teaching) in 2023.
2020 – 2023	Lecturer, Department of Information and Computing Sciences, Utrecht University, Netherlands. • Graphics for CS (and Linear Algebra)
2023 – 2024	Main Lecturer, Department of Information and Computing Sciences, Utrecht University, Netherlands. • BSc. program in Computer Science. • Logic for Computer Science. • Second half of the course, covering Inference Rules, Natural Deduction, Semantics, Hoare Logic.
2020 – 2023	AiO/TA (Lecturer), Department of Information and Computing Sciences, Utrecht University, Netherlands. • Tutorials of the Logic for Computer Science course, and Graphics Course. • Coach/Supervisor of Game Technology projects. • Supervisor of Final BSc. (Computer Science) Software Projects.
2020	Main Lecturer, University of Porto, Portugal. • Joint PhD Program in Mathematics University of Coimbra-University of Porto. • Categories in Algebra and Topology. • Second half of the course, covering Enriched Categories, Monads, and (generalized) Multicategories.
2018 – 2019	Invited Assistant Professor, Department of Mathematics, University of Coimbra, Portugal. • Topology and Linear Analysis (with Maria Manuel Clementino)
2017 – 2020	Delfos Project, Department of Mathematics, University of Coimbra, Portugal. • Taught Mathematics to secondary school students preparing for Math Olympiads.
2008 – 2012	Undergraduate teaching assistant, Universidade de Brasília, Brazil. • Differential and Integral Calculus, Real Analysis, Analysis of $\mathbb{R}^n$, Algebra 1, Linear Algebra.
2006 – 2008	Secondary Math Teacher, Curso Dínatos, Brasília, Brazil. • Lectured Secondary Mathematics to students preparing for National Higher Education Entrance Exam and Entrance Exams of various Brazilian Universities.
2005 – 2006	Student Teaching Assistant, EPCAR, Brazilian Air Force, Brazil. • Lectured Secondary Mathematics to pre-cadets.

ACADEMIC MEMBERSHIP AND SERVICE

2023 – 2025	Informatics Research Advisory Committee (IRAC) , Utrecht University. The Netherlands.
2013 – present	Centre for Mathematics, University of Coimbra , University of Coimbra. Portugal.
2020	Joint PhD Program in Mathematics , University of Coimbra and University of Porto. Portugal.

RESEARCH ASSOCIATION/GROUPS AND MEMBERSHIP

2020 – 2026	Theory of Computation , Utrecht University. The Netherlands.
2020 – 2026	Software Technology Group , Utrecht University. The Netherlands.
2013 – present	Algebra, Logic and Topology Group , Centre for Mathematics, University of Coimbra. Portugal.
2013 – present	Coimbra Category Theory Group , University of Coimbra. Portugal.

SOME RECENT AWARDS AND DISTINCTIONS

2023	Teaching Award (Distinguished Teaching) – Functional Programming , Utrecht University. The Netherlands.
2023	Fields Research Fellow , Fields Institute for Research in Mathematical Sciences. Canada.
2022	Oberwolfach Leibniz Fellow , Mathematisches Forschungsinstitut Oberwolfach. Germany.

ORGANIZATION OF EVENTS AND SEMINARS

2025 – present	Online Algebra, Logic and Topology Seminar , University of Coimbra, Portugal.
2019 – 2023	Algebra, Logic and Topology Seminar , University of Coimbra, Portugal.
2022	XIII Portuguese Category Seminar , University of Coimbra, Portugal.
2019	XII Portuguese Category Seminar , University of Coimbra, Portugal.
2018	Workshop on Algebra, Logic and Topology , University of Coimbra, Portugal.
2018	Workshop: Logic for Children , UNILOG 2018, Vichy, France.

REFEREE EXPERIENCE

Refereed for journals and conferences including, for instance, “ACM SIGPLAN Symposium on Principles of Programming Languages (POPL)”, “Journal of Pure and Applied Algebra”, “Applied Categorical Structures”, “Topology and its Applications”, “Order”, “Mathematical Structures in Computer Science”, “Quaestiones Mathematicae”, “Homology, Homotopy and Applications”, and “Theory and Applications of Categories”.

Personal Webpage

URL: <https://fllnucatelli.github.io/>

arXiv

arXiv: https://arxiv.org/a/lucatellinunes_f_1.html

Google Scholar

Google Scholar: https://scholar.google.fr/citations?user=QoW_v8QAAAAJ&hl=pt-BR

Scopus

Scopus ID 57188838707: <https://www.scopus.com/authid/detail.uri?authorId=57188838707>

OTHER LINKS

zbMath: <https://zbmath.org/authors/?q=Fernando+Lucatelli+Nunes>

Math Genealogy: <https://genealogy.math.ndsu.nodak.edu/id.php?id=228539>

nLab: <https://ncatlab.org/nlab/show/Fernando+Lucatelli+Nunes>

CMUC: <https://cmuc.mat.uc.pt/rdonweb/person/ppgeral.do?idpessoa=1331>

Portuguese School of CT: <http://www.mat.uc.pt/~categ/>